

Laboratory 2

S7001E

by

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Question 1

a) Write a short program to generate $N = 10000$ points drawn from the pdf shown in problem 3.22 in the textbook.

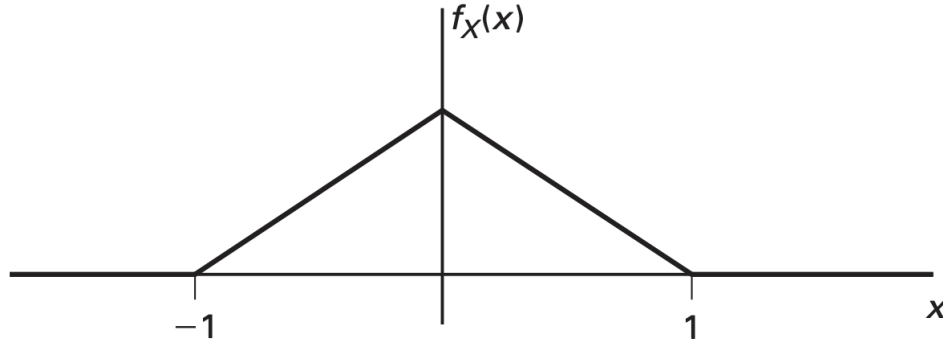


Figure P3.22

Figure 1: Figure connected to Problem 3.22 in the book [1, p. 210]

We begin by examining the figure to interpret it. We get that we can write the function for $f_x(x)$ as:

$$f_x(x) = \begin{cases} 0, & x < -1 \\ k(1+x), & -1 \leq x \leq 0 \\ k(1-x), & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

Now we need to find k . We know the area of a triangle is $A = \frac{bh}{2}$; in our case, $b = 2$ and $h = k$. We also know that for a density function it follows that the Area equals 1:

$$A = \frac{2k}{2} \implies k = 1$$

Which give us our simplified definition for $f_x(x)$:

$$f_x(x) = \begin{cases} 0, & x < -1 \\ 1+x, & -1 \leq x \leq 0 \\ 1-x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

As we have a probability density function at the moment, we want to find the CDF. The CDF is defined by:

$$F_X(x) = \int_{-\infty}^x f_x(\xi) d\xi = P[X \leq x]$$

Now we evaluate the integral over each region:

$x < -1$:

$F_X(x) = 0$ as our pdf is zero for the same interval

$-1 \leq x \leq 0$:

$$F_X(x) = \int_{-1}^x (1 + \xi) d\xi = \frac{x^2}{2} + x + \frac{1}{2} = \frac{1}{2}(x+1)^2$$

$0 \leq x \leq 1$:

$$F_X(x) = \int_{-1}^0 (1 + \xi) d\xi + \int_0^x (1 - \xi) d\xi = \frac{1}{2} + x - \frac{x^2}{2}$$

$x > 1$:

$F_X(x) = 1$ as we count the area from left to right, and it can't get bigger than 1

Now we have the full definition of $F_X(x)$:

$$F_X(x) = \begin{cases} 0, & x < -1 \\ \frac{(x+1)^2}{2}, & -1 \leq x < 0 \\ \frac{1}{2} + x - \frac{x^2}{2}, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

Now we go fourth, our object is to generate random variables from $f_X(x)$, which means that we need to find X . We can do this by letting $Y = F_X(X)$ so that we can solve for $X = F_X^{-1}(Y)$.

We do this by making expressions for x instead of $F_X(x)$ and set $F_X(x) = y$:

$-1 \leq x \leq 0$:

$$y = \frac{1}{2}(x+1)^2 \implies \pm\sqrt{2y} = x+1 \implies x = \sqrt{2y} - 1$$

This gives the interval $0 \leq y \leq \frac{1}{2}$ and it's important to notice that $\pm \implies +$ as negative would put us outside the interval for x

$0 \leq x \leq 1$:

$$y = \frac{1}{2} + x - \frac{x^2}{2} \implies x^2 - 2x + (2y - 1) = 0 \xrightarrow{pq} x = 1 \pm \sqrt{2(1 - y)} \implies x = 1 - \sqrt{2(1 - y)}$$

This gives the interval $\frac{1}{2} \leq y \leq 1$.

And now we have our X :

$$X = \begin{cases} \sqrt{2Y} - 1, & 0 \leq Y \leq \frac{1}{2} \\ 1 - \sqrt{2(1 - Y)}, & \frac{1}{2} \leq Y < 1 \end{cases}$$

The only thing left to do is to generate random numbers between 0 and 1 for Y so that we can find how X looks.

Even tho this is not included in this question, but it's easy to look if we did right by generating a normalized Histogram for X :

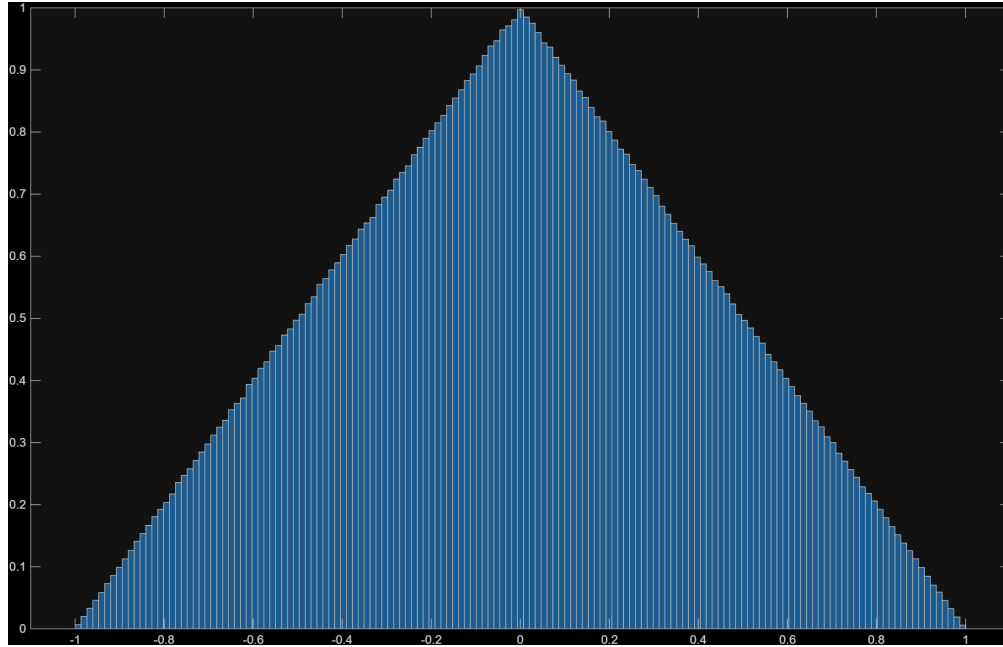


Figure 2: Histogram of X

Figure (2) is made from $N = 10\,000\,000$ points and this clarifies that we did it right.

b) Plot Histogram and theoretical pdf

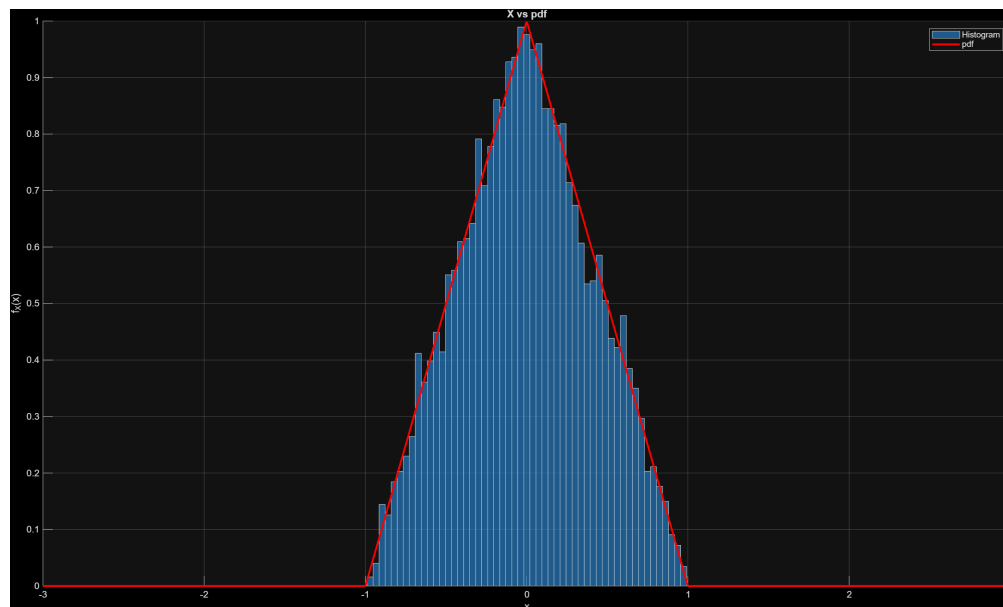


Figure 3: Histogram of X and theoretical pdf for $N = 10\,000$

The Histogram is normalized to imitate the theoretical pdf. We see in Figure (3) that they match good even though we only have 10 000 points

Question 2: Perform problem 3.23

Problem 3.23 from the book consists of generating zero-mean Gaussian numbers by adding 12 uniformly distributed numbers and subtracting 6 from the sum [1, p. 210]

We do this by generating 10.000 numbers from this procedure and producing a Histogram of the result.

This was done by randomizing a 10000×12 matrix and then summing each row and subtracting 6 from the sum. This gives us the Histogram:

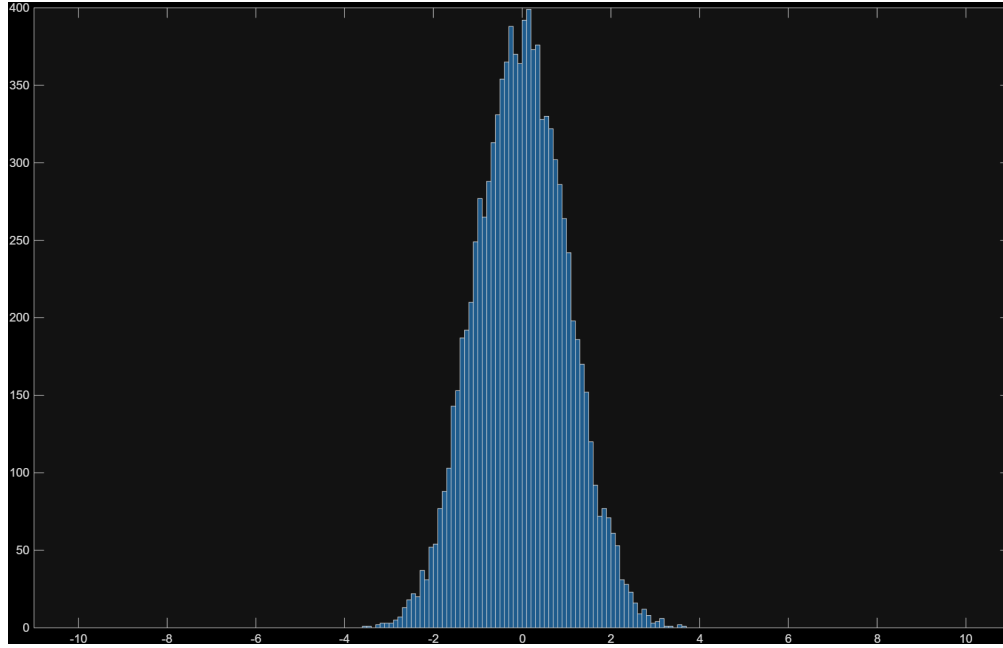


Figure 4: Histogram of our Gaussian numbers

We see that the histogram looks most Gaussian around the center.

We know that from the book that $Z = X_i + \dots + X_n$ will be approximately Gaussian for a reasonably large n , around $n > 10$ [1, p. 281]. We know that the mean value for the uniformly distributed values is $\frac{1}{2}$, which means that when we sum together 12 of these random numbers, we get a mean value of 6, and therefore the reason to why we subtract 6. If we were to increase the number of points N we would get closer to a Gaussian distribution, which we already see the same behavior in Figure (4)

Question 3: Write a Matlab program to generate $N = 10000$ points drawn from a Gaussian distributed $X \sim N(3, 2)$

This was done by first generating 10 000 Gaussian-distributed numbers, but this gives a mean of 0 and a variance of 1. so we take our Gaussian-distributed numbers X , add our mean 3, and multiply X by σ from our variance $\sigma^2 = 2$

So that we can get $X \sim N(3, 2)$:

$$X = 3 + \sqrt{2} \times X$$

a) Generate a histogram as an estimate of the pdf $f_X(x)$

Now that we have $X \sim N(3, 2)$ we can produce the Histogram for the pdf:

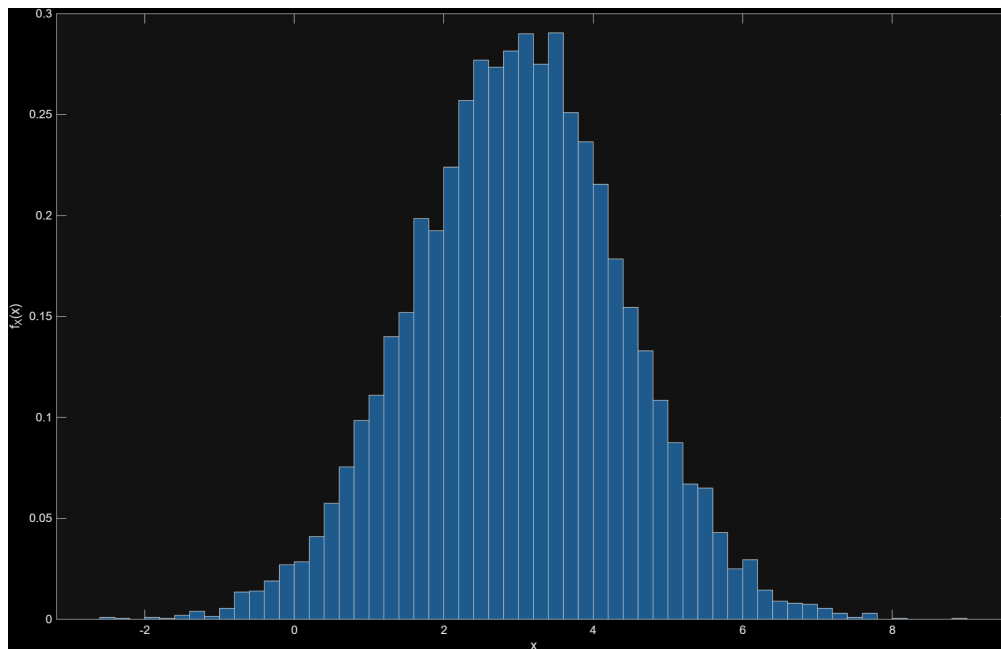


Figure 5: Histogram of $f_X(x)$

b) Try a variety of values for N and the number of bins in the histogram. Note any observations.

This was done by doing a for-loop where we did four plots with the number of points (N) and bins:

$$N = [1000, 100\ 000, 1\ 000\ 000, 10\ 000\ 000]$$

$$\text{bins} = [10, 20, 50, 100]$$

This gives us our four Histograms shown in Figure (6)

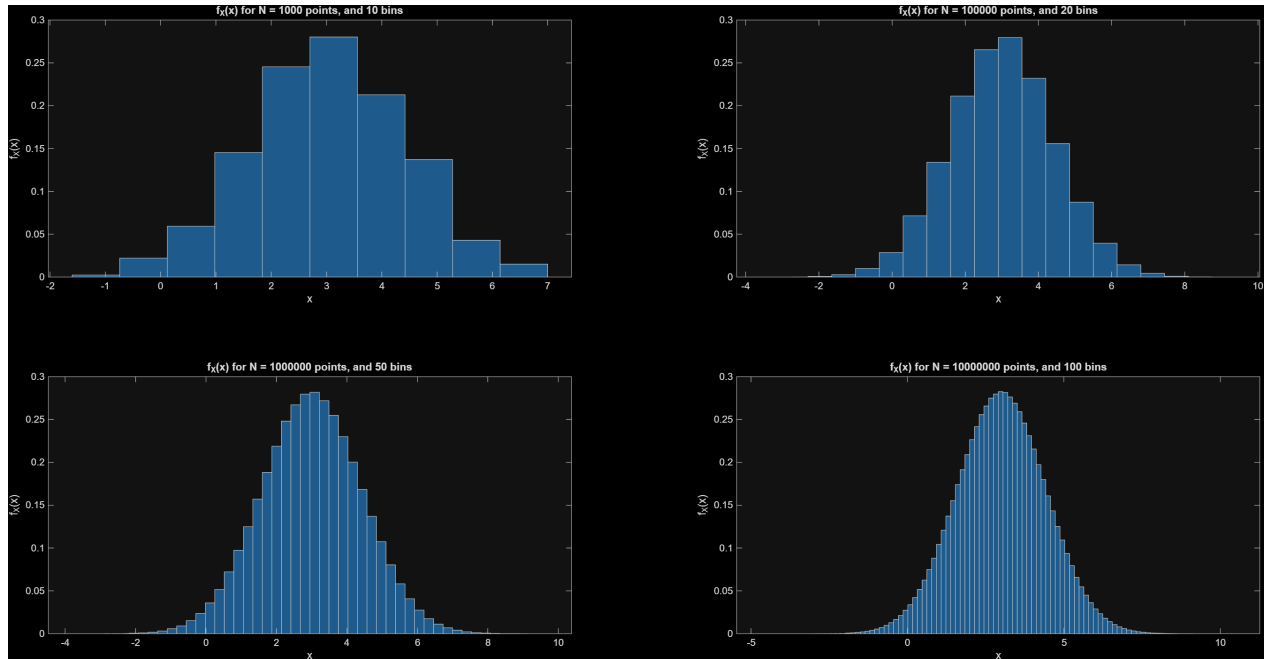


Figure 6: Histogram of $f_X(x)$ for different N and bins

We see that when we increase the number of points N and the number of bins, we get closer and closer to a perfect Gaussian distribution with its center at $x = 3$ (mean).

Question 4

We start by generating two Gaussian-distributed row vectors $X_1 \sim N(3, 2)$ and $X_2 \sim N(1, 1)$ with 10 000 points. This gives variances: $\sigma_{X_1}^2 = 2$, $\sigma_{X_2}^2 = \sigma_{X_2} = 1$ and mean values: $\mu_{X_1} = 3$, $\mu_{X_2} = 1$

a) Expression for the joint pdf $f_{X_1 X_2}(x_1, x_2)$

As we have that our row vectors X_1 and X_2 are independent of each other, it means that they are uncorrelated. which gives that $\rho = 0$.

This means that we can use $f_{X_1 X_2}(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2)$ where:

$$f_{X_n}(x_n) = \frac{1}{\sqrt{2\pi\sigma_{X_n}^2}} \exp\left(-\frac{1}{2}\left(\frac{x_n - \mu_{X_n}}{\sigma_{X_n}}\right)^2\right)$$

This comes from the book on page 251 [1].

Now we can get the expression for: $f_{X_1 X_2}(x_1, x_2)$:

$$f_{X_1 X_2}(x_1, x_2) = \frac{1}{2\sqrt{\pi}} \exp\left(-\frac{1}{2}\left(\frac{x_1 - 3}{\sqrt{2}}\right)^2\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x_2 - 1}{1}\right)^2\right)$$

Which is the same as:

$$f_{X_1 X_2}(x_1, x_2) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{(x_1 - 3)^2}{2} + (x_2 - 1)^2\right)\right)$$

We already know the variances and mean values of our row vectors as described under the section Question 4. But for the correlation coefficient, we need to first find the covariance, which we already have, as our row vectors are independent of each other:

$$\text{Cov}(X_1, X_2) = 0$$

Then the correlation coefficient will be zero as well $\rho(X_1, X_2) = 0$

b) mean values, correlation coefficient and covariance

We start by finding the means by using the MATLAB function `mean()` which can give: $\mu_{X_1} = 3.0023$, $\mu_{X_2} = 0.998$

Then we want to find the sample covariance by using the function `Cov()`, as we do this for two row vectors, we get the covariance between them returned as a 2-by-2 covariance matrix:

$$\text{Cov}(X_1, X_2) = \begin{bmatrix} 1.9661 & 0.0084 \\ 0.0084 & 0.9798 \end{bmatrix}$$

As we read MATLAB's documentation, we see that the diagonal is the variances, which means that our Covariance is along the off-diagonal: $\text{Cov}(X_1, X_2) = 0.0084 \approx 0$

For the correlation coefficient, we use MATLAB's function `corrcoef()` which returns a 2-by-2 matrix where the diagonal is ones and the correlation coefficients are along the off-diagonal.

$$\text{Corrcoef}(X_1, X_2) = \begin{bmatrix} 1.0000 & -0.0202 \\ -0.0202 & 1.0000 \end{bmatrix}$$

Which means that we get the correlation coefficient: $\rho(X_1, X_2) = -0.0202 \approx 0$

It's important to remember that the values above are taken from different runs, and that they are much closer to the values that they are approximated to when increasing the number of points.

c) Scatter plot of X_1 vs. X_2

Creating a scatter plot of X_1 vs. X_2 gives the figure:

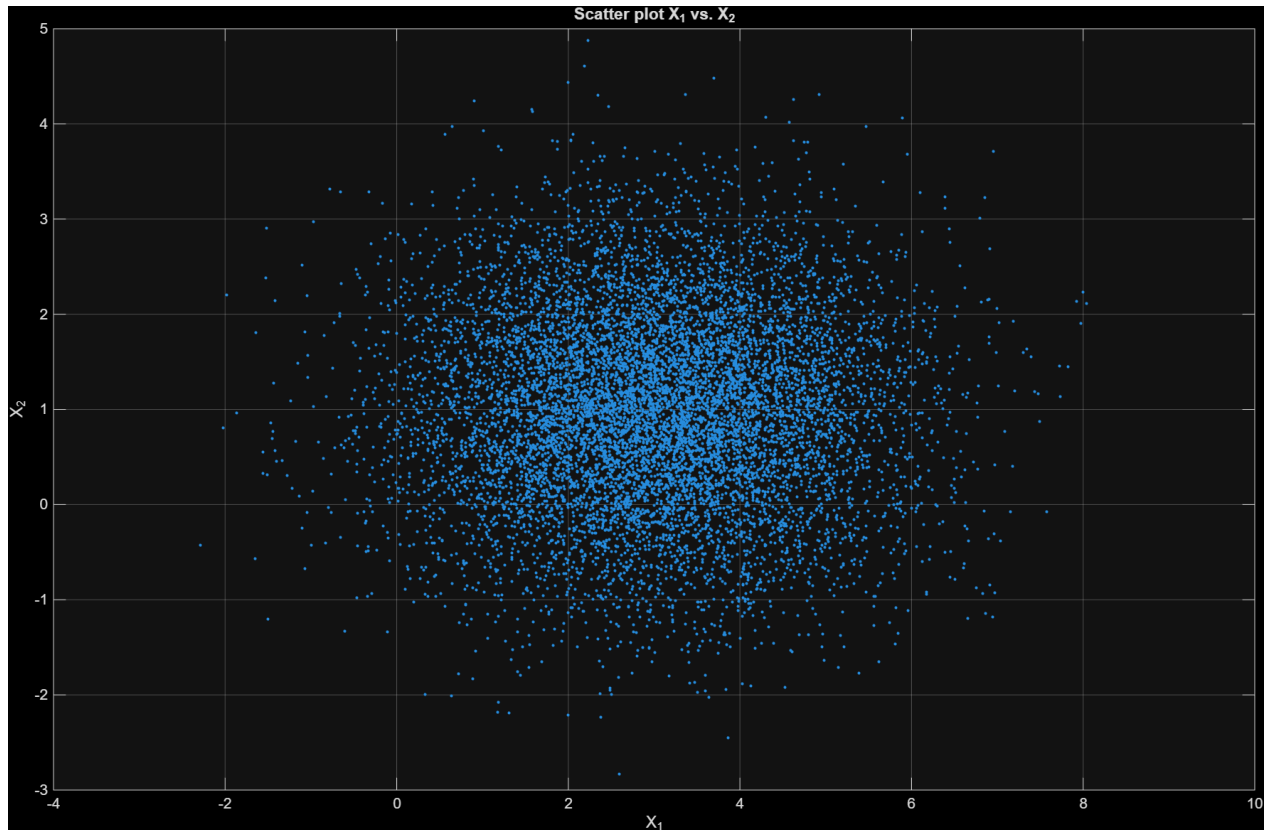


Figure 7: scatter plot of X_1 vs. X_2

From Figure (7) we can see that we got a somewhat circular shape, which is most dense in the middle, and as we go outward, it gets less dense. We can relate this to Figure 2.21 from lecture 4, given in Figure (8) below:

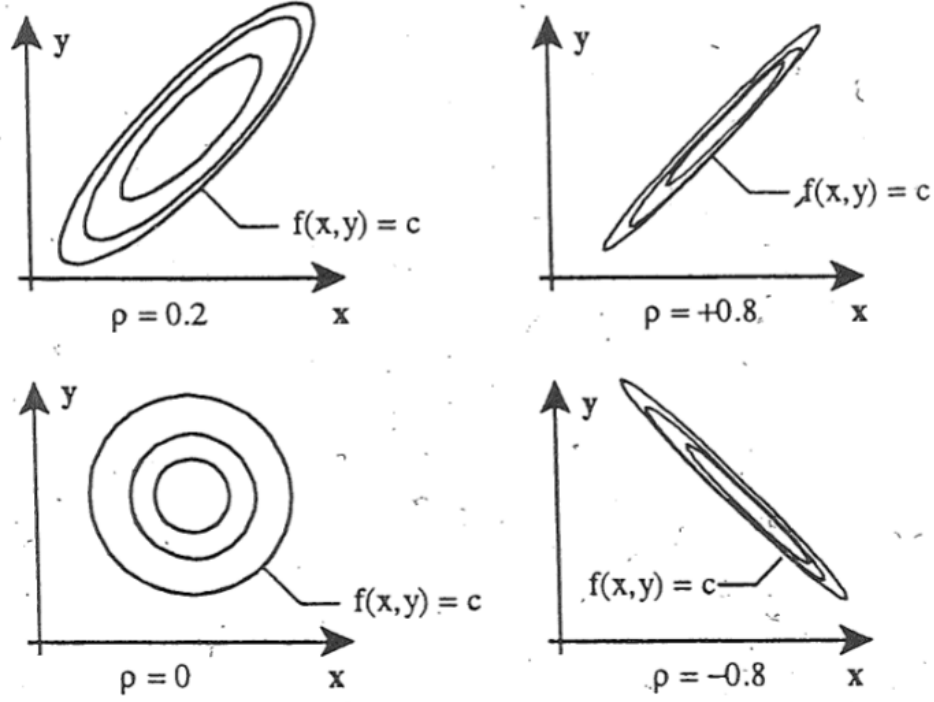


Figure 8: Contours of constant probability density functions for various values of correlation coefficients

This shows that our result from Figure (7) relates to when we have the correlation coefficient $p = 0$, which reflects the correlation coefficient we got in part b)

d) repeat a), b) and c) for Y_1 and Y_2

We have given that $Y_1 = \frac{1}{2}X_1 + 3X_2$ and $Y_2 = X_1 + X_2$. And we can start by finding the mean values:

As we know the mean values for X_1 and X_2 we can make the assumption that the mean value for Y_1 and Y_2 will be given by:

$$\mu_{Y_1} = \frac{1}{2}\mu_{X_1} + 3\mu_{X_2} \implies \mu_{Y_1} = \frac{3}{2} + 3 = 4.5$$

$$\mu_{Y_2} = \mu_{X_1} + \mu_{X_2} \implies \mu_{Y_2} = 3 + 1 = 4$$

After checking in MATLAB, we see that these mean values are correct, as they reflect the sample mean values

To find the variance, we will try to find the covariance matrix K_{YY} for Y , which means we use the formula [1, p. 323, eq. 5.6-19]:

$$K_{YY} = A_{mn}K_{XX}A_{mn}^T$$

We have that K_{XX} is the covariance matrix for X , which gives that:

$$K_{XX} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Then we have that the A matrix comes from: $Y = AX$ which is:

$$A_{mn} = \begin{bmatrix} \frac{1}{2} & 3 \\ 1 & 1 \end{bmatrix}$$

which gives:

$$A_{mn}^T = \begin{bmatrix} \frac{1}{2} & 1 \\ 3 & 1 \end{bmatrix}$$

Then we can solve for K_{YY} :

$$K_{YY} = \begin{bmatrix} \frac{1}{2} & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 9.5 & 4 \\ 4 & 3 \end{bmatrix}$$

This gives the variances $\sigma_{Y_1}^2 = 9.5$, $\sigma_{Y_2}^2 = 3$. And that we get our covariance from the off-diagonal: $\text{Cov}(Y_1, Y_2) = 4$

Now we need to find the correlation coefficient, which we do by solving the equation:

$$\rho(Y_1, Y_2) = \frac{\text{Cov}(Y_1, Y_2)}{\sigma_{Y_1} \sigma_{Y_2}} = \frac{4}{\sqrt{9.5 * 3}} = 0.7493$$

Now the only thing that is left is to find the expression for $f_{Y_1 Y_2}(y_1, y_2)$, but we can't use the same formula as for when we found $f_{X_1 X_2}(x_1, x_2)$ as they aren't independent due to $p(Y_1, Y_2) \neq 0$

We use the following equation from the book [1, p. 323, eq. 5.6-18]:

$$f_Y(y) = \frac{1}{(2\pi)^{n/2} [\det(K_{YY})]^{1/2}} \exp \left[-\frac{1}{2} (y - \beta)^T K_{YY}^{-1} (y - \beta) \right]$$

Where $\beta = A_{mn}\mu$, where $\mu = \begin{bmatrix} \mu_{X_1} \\ \mu_{X_2} \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ So we want to solve for β :

$$\beta = \begin{bmatrix} \frac{1}{2} & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 4 \end{bmatrix}$$

Now we set everything into the equation for $f_Y(y)$

$$f_{Y_1 Y_2}(y_1, y_2) = \frac{1}{(2\pi)^{n/2} \left[\det \left(\begin{bmatrix} 9.5 & 4 \\ 4 & 3 \end{bmatrix} \right) \right]^{1/2}} \exp \left[-\frac{1}{2} \begin{bmatrix} y_1 - 4.5 \\ y_2 - 4 \end{bmatrix}^T \begin{bmatrix} 9.5 & 4 \\ 4 & 3 \end{bmatrix}^{-1} \begin{bmatrix} y_1 - 4.5 \\ y_2 - 4 \end{bmatrix} \right]$$

Which we can simplify, and we also set in that $n = 2$ since Y has 2 components:

$$f_{Y_1 Y_2}(y_1, y_2) = \frac{1}{2\pi\sqrt{12.5}} \exp \left[-\frac{1}{2} \begin{bmatrix} y_1 - 4.5 & y_2 - 4 \end{bmatrix} \frac{1}{12.5} \begin{bmatrix} 3 & -4 \\ -4 & 9.5 \end{bmatrix} \begin{bmatrix} y_1 - 4.5 \\ y_2 - 4 \end{bmatrix} \right]$$

Which simplified one more step gives:

$$f_{Y_1 Y_2}(y_1, y_2) = \frac{1}{2\pi\sqrt{12.5}} \exp \left[-\frac{1}{25} [3(y_1 - 4.5)^2 - 8(y_1 - 4.5)(y_2 - 4) + 9.5(y_2 - 4)^2] \right]$$

Now we can go on to repeat part b): The sample means obtained from a single run were: $\mu_{Y_1} = 4.5128 \approx 4.5$, $\mu_{Y_2} = 4.0003 \approx 4.0$.

Then we go on to find the sample covariance: $\text{Cov}(Y_1, Y_2) = 3.9292 \approx 4.0$

Now the only thing left from the b) part is to find the sample correlation coefficient: $\rho(Y_1, Y_2) = 0.7575$, which is very close to the theoretical correlation coefficient

The only thing left here is to create the scatter plot of Y_1 vs. Y_2 :

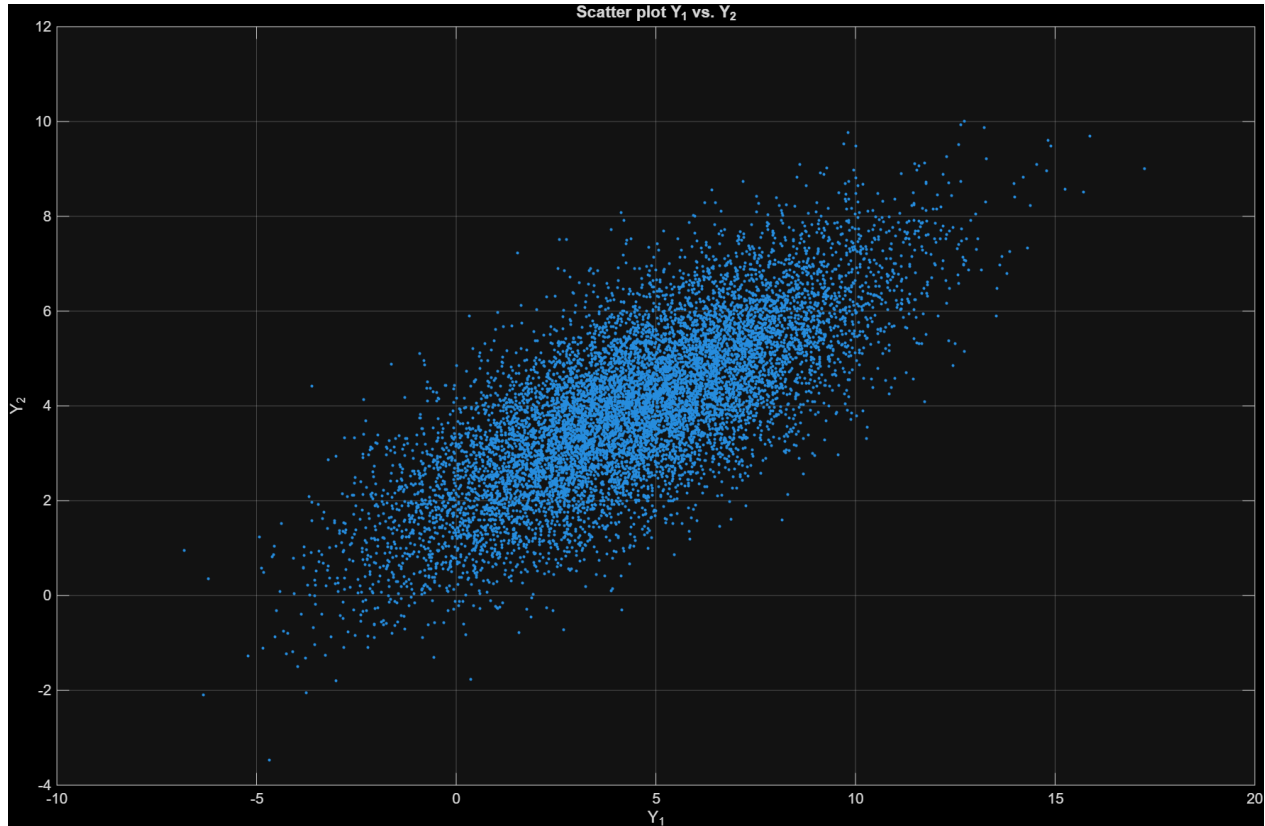


Figure 9: scatter plot of Y_1 vs. Y_2

If we compare this to Figure (8) we can see that this is pretty close to the figure for $\rho = +0.8$ where ours is a little bit bigger, which is reasonable.

e) calculate the theoretical marginal pdfs $f_{Y_1}(y_1)$, $f_{Y_2}(y_2)$

We use the same equation that we used in part a), which is from page 251 in the book [1].

$$f_{Y_n}(y_n) = \frac{1}{\sqrt{2\pi\sigma_{Y_n}^2}} \exp\left(-\frac{1}{2}\left(\frac{y_n - \mu_{Y_n}}{\sigma_{Y_n}}\right)^2\right)$$

For $Y_1 \sim N(4.5, 9.5)$

$$f_{Y_1}(y_1) = \frac{1}{\sqrt{19\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_1 - 4.5}{\sqrt{9.5}}\right)^2\right) = \frac{1}{\sqrt{19\pi}} \exp\left(-\frac{(y_1 - 4.5)^2}{19}\right)$$

For $Y_2 \sim N(4, 3)$

$$f_{Y_2}(y_2) = \frac{1}{\sqrt{6\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_2 - 4}{\sqrt{3}}\right)^2\right) = \frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(y_2 - 4)^2}{6}\right)$$

f) Expression for the theoretical conditional probability $P\{2 < Y_2 < 3 | Y_1 > 5\}$, calculate or estimate it as well

We can set $A = \{2 < Y_2 < 3\}$ and $B = \{Y_1 > 5\}$, here we see that we can do the simplification:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

So we need to find the probability of A happening at the same time as B, as well as B happening alone.

To find the probability of a joint probability density function over a certain integral/region we need to integrate the function over the space defined by the event [2].

This means that to find the probability of A and B happening at the same time, we need a double integral of $f_{Y_1 Y_2}(y_1, y_2)$:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \int_5^\infty \int_2^3 f_{Y_1 Y_2}(y_1, y_2) dy_2 dy_1$$

The probability for B is a little easier:

$$P\{Y_1 > 5\} = \int_5^\infty f_{Y_1}(y_1) dy_1$$

This give us the expression for the probability were after:

$$P\{2 < Y_2 < 3 | Y_1 > 5\} = \frac{\int_5^\infty \int_2^3 f_{Y_1 Y_2}(y_1, y_2) dy_2 dy_1}{\int_5^\infty f_{Y_1}(y_1) dy_1}$$

As these integrals can take some time to handle, we will estimate the probability of our case by using the Gaussian distribution:

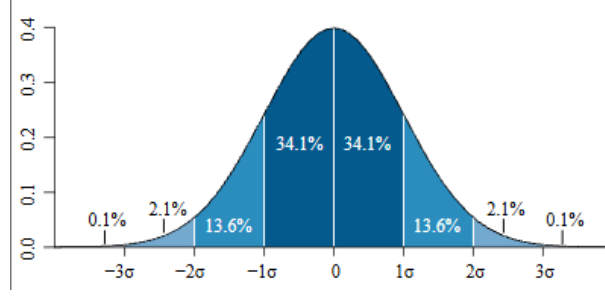


Figure 10: Gaussian Distribution (Standard deviation) [3]

For $Y_1 > 5$: $Y_1 \sim N(4.5, 9.5)$ Which gives the standard deviation $\sigma_{Y_1} = \sqrt{9.5}$

We are after finding $P\{Y_1 > 5\}$. To solve this, we will use a standard normal table, which means we need to transform to the Z-score [4]:

$$Z = \frac{X - \mu}{\sigma} = \frac{5 - 4.5}{\sqrt{9.5}} = 0.1622214211$$

Now we want to find in the table: $P(Z > 0.162)$:

$$P(Z > 0.162) \approx 0.436$$

For $2 < Y_2 < 3, Y_1 > 5$: This will get a little bit more complex, because we need to evaluate the integral, as they are not independent

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{2\pi\sqrt{12.5}} \int_5^\infty \int_2^3 \exp\left[-\frac{1}{25} [3(y_1 - 4.5)^2 - 8(y_1 - 4.5)(y_2 - 4) + 9.5(y_2 - 4)^2]\right] dy_2 dy_1$$

We can do substitution to simplify this equation:

$x = y_1 - 4.5$, $y = y_2 - 4$ This gives the new limits $0.5 < x < \infty$, $-2 < y < -1$. This gives the new integrals:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{2\pi\sqrt{12.5}} \int_{0.5}^\infty \int_{-2}^{-1} \exp\left[-\frac{1}{25} [3x^2 - 8xy + 9.5y^2]\right] |J| dy dx$$

which means we also needs to find the Jacobian J :

$$J = \frac{\partial(y_1, y_2)}{\partial(x, y)} = \det \begin{bmatrix} \frac{\partial y_1}{\partial x} & \frac{\partial y_1}{\partial y} \\ \frac{\partial y_2}{\partial x} & \frac{\partial y_2}{\partial y} \end{bmatrix} = \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

Which we can simplify some more:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{2\pi\sqrt{12.5}} \int_{0.5}^\infty \int_{-2}^{-1} \exp\left[-\frac{3}{25} \left(x - \frac{4}{3}y\right)^2\right] \exp\left[-\frac{1}{6}y^2\right] dy dx$$

We can do even one more substitution here, we are looking after to be able to use the Cumulative distribution function, which means we want it on the form: $e^{-t^2/2}$

we can extract so we have $\frac{1}{2}$ in the second exponential:

$$-\frac{1}{6}y^2 = -\frac{1}{2}\left(\frac{y}{\sqrt{3}}\right)^2$$

We do the same for the first exponential:

$$-\frac{3}{25} \left(x - \frac{4}{3}y\right)^2 = -\frac{1}{2} \left(\sqrt{\frac{6}{25}} \left(x - \frac{4}{3}y\right)\right)^2$$

Now we can do the substitution: $v = \frac{y}{\sqrt{3}}$, $u = \sqrt{\frac{6}{25}} \left(x - \frac{4}{3}y\right)$ which gives the new limits:

$$-1.154700538 < v < -0.5773502692, \sqrt{\frac{6}{25}} \left(0.5 - \frac{4}{3}y\right) < u < \infty \implies \sqrt{\frac{6}{25}} \left(0.5 - \frac{4}{3}\sqrt{3}v\right) < u < \infty$$

Where we can simplify the limit for u even more step: $\frac{\sqrt{6}}{10} - \frac{4\sqrt{3}}{5}v < u < \infty$

Now we also need to find the jacobian again:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} \frac{5}{\sqrt{6}} & 2\sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix} = \frac{5}{\sqrt{6}}\sqrt{3} = \frac{5}{\sqrt{2}} = \sqrt{12.5}$$

Putting all of this together and we get:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{2\pi} \int_{-1.1547}^{-0.5774} \int_{\frac{\sqrt{6}}{10} - \frac{4\sqrt{3}}{5}v}^{\infty} e^{-\frac{1}{2}v^2} e^{-\frac{1}{2}u^2} du dv = \frac{1}{\sqrt{2\pi}} \int_{-1.1547}^{-0.5774} \left(e^{-\frac{1}{2}v^2} \frac{1}{\sqrt{2\pi}} \int_{\frac{\sqrt{6}}{10} - \frac{4\sqrt{3}}{5}v}^{\infty} e^{-\frac{1}{2}u^2} du \right) dv$$

We see that the inner integral is a CDF:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{\sqrt{2\pi}} \int_a^{\infty} e^{-\frac{1}{2}u^2} du = 1 - \Phi(a) = \Phi(-a) = \Phi\left(\frac{4\sqrt{3}}{5}v - \frac{\sqrt{6}}{10}\right)$$

This simplifies the integral to:

$$P\{2 < Y_2 < 3, Y_1 > 5\} = \frac{1}{\sqrt{2\pi}} \int_{-1.1547}^{-0.5774} e^{-\frac{1}{2}v^2} \Phi\left(\frac{4\sqrt{3}}{5}v - \frac{\sqrt{6}}{10}\right) dv$$

From here on we can solve this equation using MATLAB's functions `normcdf()` for Φ and `integral()` to solve the whole integral. This gives the output

$$P\{2 < Y_2 < 3, Y_1 > 5\} = 0.013292583598087$$

Now we can get the probability for:

$$P\{2 < Y_2 < 3 | Y_1 > 5\} = \frac{P\{2 < Y_2 < 3, Y_1 > 5\}}{P\{Y_1 > 5\}} = \frac{0.013292583598087}{0.436} = 0.03048757706$$

e) Estimate $P\{2 < Y_2 < 3 | Y_1 > 5\}$ from generated data

Generating this from our generated data gives us the probability:

$$P\{2 < Y_2 < 3 | Y_1 > 5\} = 0.04311$$

Which means that we weren't too far away when estimating the probability in part f)

Question 5 Separation of speech signals

Let $X(t)$ and $Y(t)$ be the processes corresponding to NSD and Kuriren, respectively. We assume that these processes are mixed linearly without memory in the left (L) and right (R) channels of the received stereo sound. For simplicity, we also assume that the amount of mixing is the same for both channels, that is, the mixing may be written with the matrix relation

$$\begin{bmatrix} 1 & a \\ a & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} L \\ R \end{bmatrix}$$

With this simple model, we can find estimates of X and Y as

$$\hat{X} = L + kR$$

$$\hat{Y} = kL + R$$

where k =scalar

`newspapers.wav` is the received signal, `newspaper1.wav` and `newspaper2.wav` that are only the left and right channels, respectively.

Start with the received signal and try to separate it into the two different newspapers (better than the left and right channel) by adaptively minimizing the correlation between $\hat{X}$ and $\hat{Y}$

a) We choose to minimize the cost function $Q = \left(E[\hat{X}\hat{Y}]\right)^2$ in order to minimize the correlation. Express Q in terms of k , L and R .

We can start by finding the product of $\hat{X}$ and $\hat{Y}$:

$$\hat{X}\hat{Y} = (L + kR)(kL + R) = kL^2 + LR + k^2LR + kR^2 = k(L^2 + R^2) + (1 + k^2)LR$$

Which gives that:

$$E[\hat{X}\hat{Y}] = E[k(L^2 + R^2) + (1 + k^2)LR]$$

We can use the relationship for the operator E and how we can simplify when we have linearity [1, p. 222], and that we can take out our constants:

$$E[\hat{X}\hat{Y}] = kE[L^2 + R^2] + (1 + k^2)E[LR] = kE[L^2] + kE[R^2] + (1 + k^2)E[LR]$$

Now we can put this in to our equation for Q :

$$Q = \left(E[\hat{X}\hat{Y}]\right)^2 = \left(kE[L^2] + kE[R^2] + (1 + k^2)E[LR]\right)^2$$

b) To determine the k that minimizes Q we can find the k that gives $\frac{dQ}{dk} = 0$, express derivative in terms of k , L and R .

The derivative will give:

$$\frac{dQ}{dk} = 2E[\hat{X}\hat{Y}] \frac{dE[\hat{X}\hat{Y}]}{dk}$$

So we start by solving $\frac{dE[\hat{X}\hat{Y}]}{dk}$:

$$\frac{dE[\hat{X}\hat{Y}]}{dk} = E[L^2] + E[R^2] + 2kE[LR]$$

Putting it all together we get:

$$\frac{dQ}{dk} = 2(kE[L^2] + kE[R^2] + (1 + k^2)E[LR])(E[L^2] + E[R^2] + 2kE[LR])$$

Simplify, we do the substitution $A = E[L^2]$, $B = E[R^2]$, $C = E[LR]$ to do it easier:

$$\frac{dQ}{dk} = (2kA + 2kB + (2 + 2k^2)C)(A + B + 2kC)$$

Then:

$$\frac{dQ}{dk} = 2kA^2 + 2kAB + 4k^2AC + 2kAB + 2kB^2 + 4k^2BC + (2 + 2k^2)CA + (2 + 2k^2)CB + (2 + 2k^2)2kC^2$$

$$\frac{dQ}{dk} = (4C^2)k^3 + (6AC + 6BC)k^2 + (2(A + B)^2 + 4C^2)k + (2AC + 2BC)$$

$$\frac{dQ}{dk} = 4C^2k^3 + 6C(A + B)k^2 + 2((A + B)^2 + 2C^2)k + 2C(A + B)$$

And now we set in: $A = E[L^2]$, $B = E[R^2]$, $C = E[LR]$:

$$\frac{dQ}{dk} = 4E[LR]^2k^3 + 6E[LR](E[L^2] + E[R^2])k^2 + 2((E[L^2] + E[R^2])^2 + 2E[LR]^2)k + 2E[LR](E[L^2] + E[R^2])$$

c) Load the file

We load the file `newspapers.wav` and we get the matrix X , where the first column contains the left channel and the second column contains the right channel

d) Estimate the expected values

To say, we want to estimate $E[L^2]$, $E[R^2]$, $E[LR]$ for the first $N = 100$ samples.

We do this by calculating the mean:

$$E[L^2] = \frac{1}{N} \sum_{n=1}^N L_n^2$$

$$E[R^2] = \frac{1}{N} \sum_{n=1}^N R_n^2$$

$$E[LR] = \frac{1}{N} \sum_{n=1}^N L_n R_n$$

If we set in our samples (N), we get the expected values:

$$E[L^2] = 0.0209$$

$$E[R^2] = 0.0208$$

$$E[LR] = 0.0092$$

e) Update expected values for each sample

The d) part was only to initialize the estimation of our expected values; now we are out after estimating the expected values using the rest of the values from the left and right channels.

We will do this by iterating from the 101st sample until the last one, where, in every iteration, we keep most of the old expected value and let the instantaneous values be only a small part.

This is done with $\lambda = 0.99995$, which gives the formula for the next expected value. To say the value at iteration i depends on the expected value from the iteration before it:

$$E[L^2]_i = (1 - \lambda)L_i^2 + \lambda E[L^2]_{i-1} = 0.00005L_i^2 + 0.99995E[L^2]_{i-1}$$

$$E[R^2]_i = 0.00005R_i^2 + 0.99995E[R^2]_{i-1}$$

$$E[LR]_i = 0.00005L_iR_i + 0.99995E[LR]_{i-1}$$

This means that we keep 99.995% of the earlier value and add 0.005% of our new value.

It's important to note here that we will start at index $i = 101$, so we don't take data from the left and right channels, which we used for our initial estimation of the expected values.

This gives us our new estimated values:

$$E[L^2] = 0.0091$$

$$E[R^2] = 0.0065$$

$$E[LR] = 0.0036$$

f) Estimation of constant k

We are out after estimating our constant k , this is done with the following equation:

$$k = k - \mu \times \text{sign} \left[\frac{dQ}{dk} \right]$$

where μ is our step size which is set to be $\mu = 10^{-5}$, we also have the starting value $k = 0$.

This exercise was solved by putting the equation for k in to the same loop we used in part e). and also our definition for $\frac{dQ}{dk}$ from part b). Which means we redesigned k as:

$$k_i = k_{i-1} - \mu \times \text{sign} \left[\frac{dQ}{dk} \right]$$

This give ous the value for k which minimizes Q :

$$k = -0.2401$$

We put this value into our expression for $\frac{dQ}{dk}$ from part b):

$$\frac{dQ}{dk} = 1.588 \times 10^{-6} \approx 0$$

Which means we have succeeded with our value for k that minimizes Q

g) Store the value for k at each iteration in a vector, so we can estimate $\hat{X}$, $\hat{Y}$. Plot k-vector

This was solved by creating an empty vector for our k -values and saving them for each iteration in the same loop as we used above, we can call the vector as k_{values} .

And then we calculated $\hat{X}$ and $\hat{Y}$ with the updated formulas:

$$\hat{X}_i = L_i + k_{\text{values}, i} R_i$$

$$\hat{Y}_i = k_{\text{values}, i} L_i + R_i$$

And then we have the plot of how k changes for each iteration:



Figure 11: plot of k_{values}

h) Write the separated signals to wav-files

After saving our new wave files and comparing them to the ones on Canvas, we can hear a big difference. The new ones have much clearer audio, where the background audio from the other channel is much quieter and dampened. This applies to both wave files.

My theory is that the background noise doesn't disappear completely, as we don't minimize Q completely. i.e. $\frac{dQ}{dk} = 0$. You could try to optimize the estimation of k to obtain a better result.

References

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